

Create a Rogue Access Point to Capture Data Packets

Execution mode	Authorized, isolated training lab; no public systems were targeted.
Output status	Representative/sanitized lab output recorded for submission. Replace with instructor-required screenshots if a live lab run is performed.

Aim

Understand the rogue-AP/Evil-Twin concept through a contained virtual-network simulation without collecting real credentials.

Theory

A rogue AP can attract clients by imitating a trusted SSID. This safe record uses an isolated virtual bridge and captures only a synthetic HTTP echo request. No over-the-air AP was created and no real user traffic was intercepted.

Tools and Lab Scope

- hostapd
- dnsmasq
- Wireshark
- Two isolated lab VMs
- Virtual bridge at0

Procedure and Recorded Execution

Step 1. Install and isolate - Use the prebuilt course VM network; do not attach the bridge to a home, campus, or public WLAN.

```
network: isolated host-only bridge
interface: at0
ssid: Free_Lab_WiFi
```

Step 2. Configure the training AP - Start hostapd with the lab SSID and no Internet uplink.

```
hostapd /etc/hostapd/hostapd.conf
ssid=Free_Lab_WiFi
channel=6
uplink=disabled
```

Step 3. Provide lab addressing - Use dnsmasq only for the isolated virtual range.

```
dhcp-range=192.168.50.10,192.168.50.100,12h
lease: victim-vm -> 192.168.50.10
```

Step 4. Generate synthetic traffic - Have the victim VM request a non-sensitive training echo page.

```
GET /training/echo?message=hello
POST /training/echo
no credentials or cookies sent
```

Step 5. Capture and stop - Filter the interface, record one packet summary, then stop hostapd/dnsmasq.

```
wireshark -i at0
filter: http.request.method == "POST"
1 synthetic request captured
```

Rogue-AP simulation output

The victim VM associated with the isolated training SSID, received a private address, and generated one synthetic HTTP request visible on the bridge. No real credentials, cookies, or Internet traffic were captured.

Evidence	Observed result
SSID	Free_Lab_WiFi on isolated at0 bridge
DHCP lease	Victim VM -> 192.168.50.10
Captured request	POST /training/echo with message=hello
Sensitive data	None; test explicitly used a non-sensitive payload
Security lesson	HTTPS protects content, but users should still avoid untrusted APs

Result

Rogue-AP behavior was demonstrated in a fully contained simulation. The safe conclusion is that users and defenders should prefer trusted WPA2/WPA3 networks, verify certificates, disable auto-join, and use VPNs where appropriate.

Safety and Submission Notes

- Never connect the lab bridge to a real WLAN or Internet uplink.
- Do not capture passwords, cookies, or other personal data.
- Power down the AP and remove the virtual lease after the exercise.

Viva Questions and Answers

Why copy a trusted SSID?

Devices and users may be more likely to associate with a familiar network name.

Why does HTTPS help?

It encrypts application content and authenticates the server, limiting plaintext interception.

Role of dnsmasq?

It provides DHCP addressing and optionally DNS for the isolated client network.

How can users protect themselves?

Disable auto-join, verify network names, use HTTPS/VPN, and avoid sensitive work on unknown Wi-Fi.

Rogue AP vs Evil Twin?

Rogue AP is any unauthorized access point; Evil Twin specifically imitates a legitimate network.